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CropSyst model evolution: From field to regional to global scales and from research to decision support systems☆

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ABSTRACT

Motivated by global and regional challenges in food production and a broader consideration of ecosystem services, there has been a substantial increase in the demand for integrated agricultural systems models and spatially-distributed applications that can be used for regional and global assessments and as decision support tools. This demand marks a shift from earlier emphasis in single-crop point simulations and poses a significant challenge as cropping systems models need to improve and increase their capabilities to address multiple scales in a cohesive scientific manner and using updated computing platforms. In this article we discuss how the cropping systems model CropSyst has evolved to meet these new demands and provide some concepts for the future.

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1. Introduction

The 2003 special issue of Agricultural Systems was devoted to the evaluation of the state-of-the-art of cropping systems models (Van Ittersum and Donatelli, 2003) including CropSyst (Stöckle et al., 2003). This publication was timely, coinciding with the ending of a period where the description and testing of single crop models was becoming a less attractive line of work for publication in relevant peer-reviewed journals.

Although a large number of single-crop models exist, for example the study of Asseng et al. (2013) includes 27 wheat models (many other exist that specialize on maize, rice, and other crops), the number of cropping system models (CSMs) is much smaller. Examples include DSSAT (Jones et al., 2001), APSIM (McCown et al., 1996), EPIC (Williams et al., 1984), STICS (Brisson et al., 2003), CropSyst (Stöckle et al., 2003), and a few others. A description and comparison of the modeling solutions of these different models is beyond the scope of this article, but suffice to say that history, resources, modeling philosophies, and software development decisions have led to distinctive characteristics of these models while available literature on biophysical, physiological and other scientific understanding of crop growth processes in response to the environment have provided many elements of commonality.

Today, complex global change drivers require the study of agroecosystems at different temporal and spatial scales and with higher integration of diverse scientific disciplines and decision-making issues. This reality is creating challenges and opportunities for CSMs, resulting in a growing interest in the development of software that facilitates the implementation of integrated modeling frameworks with multiple and diverse processes, disciplines (e.g., agronomy, hydrology, biophysics, geochemistry, economics, and social sciences), scales, and opportunities for stakeholders input (Kelly et al., 2013). This trend contrasts with the predominant approach of using standalone CSMs as single tool for the evaluation of weather and management impacts on crop yield and the environment at the field scale. In some instances, the use of these models has been extended from field to larger scales through links to geographical information systems, where landscape units are not hydrologically connected (Hartkamp et al., 1999; Priya and Shibasaki, 2001).

As pointed out a decade ago, models are rarely designed to communicate with other models within a discipline, and even less with models from other disciplines (Argent, 2004). Cropping system models have been used in interactive mode with another

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model (e.g., an economic model) via manual output–input link or by using simple interface programs (e.g., Finger, 2013). Examples of greater degree of software integration of CSMs with other models exist but they are limited (e.g., Negm et al., 2014; Arnold et al., 2010). Multi-model ensembles are also emerging, where many models are linked for specific purposes. For example, the BioEarth project (Adam et al., 2014) explores the coupling of multiple stand-alone models integrating atmospheric, terrestrial, aquatic, and economic models to generate usable information for agricultural and natural resource decision-making at the regional scale.

The continuing demand for higher levels of model integration to address complex environmental problems is expected to lead to the assemblage of software platforms that include modeling solutions from multiple disciplines instead of coupling whole stand-alone models (Donatelli et al., 2014). These platforms may include modules and processes extracted from existing CSMs and models from other disciplines within a framework that can be applied at different spatial and temporal scales and implemented with economic, social and stakeholder interaction capabilities.

It appears unavoidable that cropping systems models applicable to the study of large-scale land use and environmental problems will have to evolve towards new paradigms of higher integration. In this article we discuss the recent evolution of CropSyst as first steps towards adaptation to this emerging reality, and which is presented as an example of the challenges being faced by similar cropping systems models.

2. Evolution of CropSyst simulation approach

The simulation of crop growth in response to the surrounding environment and management is central to cropping systems models. The crop modeling approach in CropSyst has remained distinctive and consistent throughout the model's evolution. Crop growth is simulated using a resource-capture and resource-use efficiency concept, where the main resources are radiation, water, and nutrients. Resource capture is modulated by phenology defining the duration of the growing season, and canopy and root growth and architecture. All these are simulated based on biophysical principles or observable processes that lend themselves to empirical parameterization. Details on process modeling are given elsewhere and will not be discussed here (e.g., Jara and Stöckle, 1999; Stöckle and Debaeke, 1997; Stöckle et al., 2003; Kemanian et al., 2005, 2007; Kremer et al., 2008; Stöckle et al., 2010).

As has been the experience of other cropping systems models (e.g., DSSAT, EPIC), new demands on agro-ecosystem simulation capabilities have led to further development of CropSyst to account for the combined cycling of carbon and nitrogen, including nitrous oxide emissions from nitrification and denitrification (Stöckle et al., 2012). This has also led to complementary modules to calculate carbon footprint of agricultural systems based on a life cycle assessment approach (Zaher et al., 2013). Other model enhancements in the last 10 years include improvements in the use of water-use efficiency as driver of biomass production (Kemanian et al., 2005; Kremer et al., 2008), improved approaches to account for elevated atmospheric CO_2 on biomass production (Stöckle et al., 2010), and better accounting of extreme heat on harvest index and grain yield (unpublished). In addition, interest in large-scale regional analyses have brought about the need to simulate a larger number of crops. The generic crop growth framework of CropSyst has proved suitable to simulate not only annual and perennial herbaceous crops, but also horticultural and fruit tree crops (Oyarzun et al., 2007; Marsal and Stöckle, 2012). A prototype version is under development to simulate greenhouse horticultural crops (Gallardo et al., 2012).

Perhaps the most significant development of the last 10 years has been an increased demand for the use of cropping system models to assess climate change impact on agriculture and evaluate avenues for mitigation and adaptation, conducted at ever larger spatial scales and degrees of agricultural systems complexity. CropSyst capabilities have been upgraded, and improvements to the model are a continuous process to meet these needs. Incorporation of CropSyst into spatially-distributed hydrologic models and grid-based regional simulation platforms, and of selected elements of the model into specialized tools for analysis or decision support, are becoming of increasing interest as the rapid progress of hardware and software capabilities is opening the door for innovation as not seen before. Examples of these applications are discussed below.

3. Applications to climate change impact and mitigation assessment

Driven by ensuing climate change and increasing atmospheric concentration of carbon dioxide, there is renewed interest on crop model applications to evaluate impacts on agricultural productivity. The extent to which agriculture can mitigate climate change via reduced emission of greenhouse gases or sequestration of carbon is also of interest. Concerns about climate change impact on agricultural water use and competing demand from other users need to be addressed.

Simulation models are essential tools to evaluate strategies to assess these challenges.

3.1. Case study applications

Case studies focus on selected locations to draw conclusions and shed light into processes and system responses that could be generalized to larger regions. Stöckle et al. (2010) have presented a typical CropSyst case study application using selected locations and crops to conduct a preliminary assessment of climate change impacts on agriculture in Washington State, USA including some simple adaptations (earlier planting of spring/summer crops and the use of slower-maturing cultivars). Other examples include Tubiello et al. (2000), Abraha and Savage (2006), and Sommer et al. (2012). In south-eastern Australia, CropSyst simulations showed wheat yield reductions despite simple adaptations, which must be counteracted by advances in agronomy and breeding to keep pace with projected climate change (Anwar et al., 2007). Climate mitigation via increased soil organic carbon (SOC) storage and reduced greenhouse gas (GHG) emissions in agriculture may help curve the current increasing rate of atmospheric CO_2 concentration. Stöckle et al. (2012) and Zaher et al. (2013) used CropSyst to assess the potential for SOC storage and changes in GHG emissions resulting from the conversion from conventional tillage to reduced or no-tillage management in Washington State, USA.

Crop water use under climate change has become a research area of great interest. For example, CropSyst was used to assess the impact of climate change scenarios on the water footprint of maize in the Po valley of Italy (Bocchiola et al., 2013) and to estimate vineyard full-irrigation requirements under climate change scenarios for the Ebro Valley, Spain (Marsal and Utset, 2008). In other studies, CropSyst was included in a bio-economic modeling framework at the level of a water users association in Uzbekistan (Sommer et al., 2010, 2011, Djanibekov et al., 2013) and in an economic model that represents the decision-making process of risk-averse farmers to assess the sensitivity of agricultural water use to price variability and climate change (Finger, 2012).

3.2. Applications at regional scales

The need to assess climate change impacts on agricultural systems at regional and global scales to inform policy makers on issues such as food supply and water resource allocation have opened a new direction for model applications where their use is essential. To respond to these demands, there is a need to use cropping system models with a large array of management options (e.g., tillage, fertilization, irrigation, crop rotations, planting dates, seeding rates, cultivar selection), and able to simulate water, carbon, and nitrogen budgets and cycling as a minimum. We review here a few applications of CropSyst for regional assessment of climate change impact on agriculture.

Sommer et al. (2013) evaluated the response to climate change and elevated atmospheric CO_2 concentration of 14 wheat cultivars grown with three levels of agronomic management in 18 sites in agro-ecological zones of Kazakhstan, Kyrgyzstan, Uzbekistan, and Tajikistan. This is an interesting and unusual study given the wide range of cultivars and management scenarios included. The climate change impact in the Mediterranean basin at the time of a 2 °C warming was examined by Giannakopoulos et al. (2009) using gridded weather data with a resolution of 2.5° of latitude by 3.75° longitude, including 13 grid cells ("hot spots") where changes in precipitation and temperature patterns are expected to be substantial. Other CropSyst-based regional studies include an assessment of crop yield for selected crops (Bambara nut, groundnut, soybean, maize and sorghum) in eight agricultural regions in Cameroon (Tingem and Rivington, 2009), an impact assessment of extreme climate events on grain yields (Moriondo et al., 2011), an evaluation of effects of changes in mean climate and climate variability on the yield of winter and spring crops in Switzerland (Torriani et al., 2007), and the assessment of climate change and price scenarios on maize production at the Swiss Plateau (Finger et al., 2011).

In another study, a combination of remote sensing data, climate data, and CropSyst simulations were applied at a grid scale of 25 km × 25 km to assess the spatial variability of cropping system response to climate change in Punjab State, India (Tripathy et al., 2011). Also, the integration of CropSyst into an Agro-ecological Zones Simulator at a resolution of 25 km × 25 km and its application for the assessment of climate change-productivity impact in Latin America and the Caribbean was reported by Confalonieri et al. (2012). In this study, in addition to the direct effect of climate change on yield of maize, the effect of biotic stress (in this case, gray leaf spot disease) was also simulated, an important trend for future studies of climate change impacts.

4. CropSyst platform for grid-based regional assessments

For the assessment of agricultural systems at regional and global scales, the use of grid-based simulations is important (Liu, 2009; Japtag and Jones, 2002). In the US Pacific Northwest, CropSyst is being used as part of a large interdisciplinary effort (REACCH project; https://www.reacchpna.org/) to study climate change impacts on current and future wheat-based cropping systems, including effects on productivity, SOC, GHG emissions, and water and nitrogen budgets. The region, comprised of portions of Washington, Oregon, and Idaho, is divided in grid cells of 4 km × 4 km, with weather provided by downloads from 14 GCMs considering two future concentration pathways of atmospheric CO_2.

This greater complexity necessitates significantly more computational resources. The CropSyst framework has been upgraded to run simulations under platforms other than MS-Windows desktops such as Linux-based high-performance computing (HPC) clusters and supercomputers.

For grid-based projects, simulation points are defined by the centroids of the weather grid cells. The grid is derived from downscaled GCM weather data, and cell dimensions are determined by the downscaling procedure. There is no requirement that land units be spatially oriented in a grid. The spatial topology could also be polygons or specific points. The weather and soil maps (either grid or polygons) are spatially joined using the map overlay function of GIS software. Other distinguishing properties of a land unit could be included in this joining procedure such as crop rotation and management regimens. Ultimately a table is created that identifies homogeneous land units to be simulated.

HPC systems use a resource batch scheduler to automatically initiate invocations of programs on available nodes/processors. Most Linux systems use a scheduler that can be used not only on large cluster systems, but also on smaller multicore desktops or servers. A macro is written to generate batch scripts to run the model for each homogeneous land unit identified by the overlaid map. The framework uses the operating system's hierarchical directory file structure to organize the simulation run inputs and outputs. Subdirectories may correspond to weather, future atmospheric CO_2 concentration pathways, cropping system scenarios, or other significant categories. The directory hierarchy can be structured dynamically as needed or convenient. Since one may be using multiple servers and HPC clusters the directory structure facilitates distributing job workload to the distributed servers. It also helps to quickly find outputs where a directory can serve as a point where outputs can be tallied and/or aggregated. The software that supports grid-based simulation runs of CropSyst will be eventually available for download along with the existing access to the CropSyst Suite.

5. Integration of CropSyst into specialized tools

CropSyst, either as a complete system model or its constituent components, has been integrated into standalone applications, modeling suites, and decision support systems. For example, Bechini and Stöckle (2007) integrated a MS Excel version of CropSyst and a regional data base (MS Access) to produce a tool for simple farm-scale analyses. Bechini et al. (2008) expanded this concept and developed a computer program that integrates CropSyst with several regional agricultural databases and provides the users with a simple framework to support the preparation and evaluation of nitrogen fertilization plans for livestock farms in the Lombardy region, northern Italy. Other ongoing projects include the Organic Farming Footprint Tool (OFoot) and a dairy farm nutrient management system, which are described below.

5.1. Organic Farming Footprint Tool (OFoot)

Many organic farmers aim at providing a service to their environment by utilizing sustainable and responsible farming practices. OFoot (http://csanr.wsu.edu/organic-farming-footprints/) attempts to quantify these environmental services by allowing users to estimate carbon footprints for farms and farm products. OFoot estimates carbon footprints through a web-based tool that integrates CropSyst with farm management software and a carbon footprint calculator.

OFoot is a rich interactive web application that is built upon a collection of open source code. The technologies used to develop the tool can be related to the four components of the tool; the View, Database, Models, and Controller (Fig. 1). The View handles representational logic, user-interface components and data mapping from the Database to the View. The Database allows data

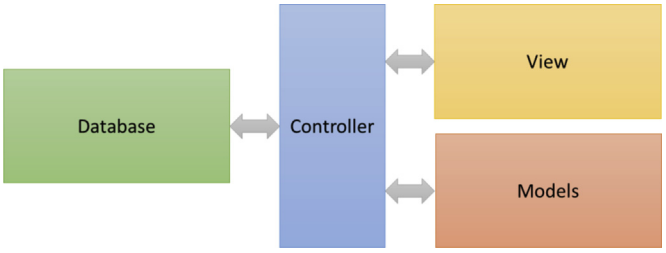


Fig. 1. Component of the OFoot tool and their interactions. The Database stores application data. The controller contains communicative logic. The View contains representational logic. The Models, which include CropSyst, are necessary to calculate a carbon footprint from data in the Database.

persistence and acts as a data-store for a suite of pre-defined model parameters. The Models, which ultimately convert data from the Database into a carbon footprint, include CropSyst and a carbon footprint calculator (CFCalc). Finally, the Controller handles the communication logic between the Database, the View, and the Models.

The OFoot user-interface collects data related to the farm and to crop management practices. Data collection can be broadly divided into two paradigms that also function to give two resolutions of an estimated carbon footprint (Fig. 2). The first paradigm, the Farm Inventory, provides a broad estimation of a farm's carbon footprint by obtaining an annual inventory of the farm's infrastructure, equipment, and consumables. The second paradigm, the Field Simulation, gives a finer resolution carbon footprint for farm products by obtaining details about farm management practices on a given area of land. The user must define a series of typical farm operations for one or more crops (crop management), then define a typical sequence of crop plantings (cropping sequence), and finally give operational details including the type and amount of consumables used (operation details).

A CropSyst scenario is generated by OFoot using data from the Database. User input defined in the Cropping Sequence and Crop Management aspects of the View are used to generate a CropSyst rotation file and a collection of CropSyst management files, respectively. Crop and organic amendment parameter files are selected from the Database based on user input in the Operation Details section of the View. Finally, the geolocation of the field is used by CropSyst to query a local spatial database for weather and soil data.

Results from the CropSyst scenario are passed to CFCalc to generate a final carbon footprint for the farm products. Soil carbon flux is estimated using a three pool carbon model (Stöckle et al., 2012). Soil N_2O emissions is estimated as part of a nitrogen cycling submodel. Finally, water used for irrigation is estimated based on simulated demand needed for economic crop growth. CFCalc converts these values to CO_2 equivalents and assigns them, using a weighted average, to the different crops specified in the cropping sequence. The result is a carbon footprint for each farm product in the simulation.

5.2. Dairy nutrient management

CropSyst is being integrated into new software development aimed at evaluating the carbon footprint and groundwater pollution associated with alternative management and technologies of a dairy farm operation. The dairy manure handling model (Uslar, 2010) estimates the production, emission and losses of nitrogen, phosphorus, and carbon from animals and manure at different stages of manure treatment in a dairy operation. A typical dairy operation is divided in components including animal, barn, solid separator, lagoon, anaerobic digester, nutrient recovery system, pyrolysis, and land application, which may or not be present in specific scenarios (Fig. 3). Tracking nutrients in dairy manure during storage and handling allows users to estimate the manure characteristics before and after each manure treatment, eventually defining effluent characteristics to be handled by CropSyst during crop land application. Outputs of the model are unit operation specific or whole system quantification of GHGs and other gaseous emissions, mass/energy balances, nutrient flows and transformations, and impacts on crop yield and SOC storage.

6. Integration of CropSyst into hydrologic models

To address issues of agricultural water supply and demand at basin scales, CropSyst has been coupled with the Variable Infiltration Capacity (VIC) model (Liang et al., 1994). In addition, a hydrologic model for small agricultural watersheds, CropSyst-Microbasin, is under development for small-watershed applications such as understanding the effect of topography and cropping systems management on water and nutrient distribution, identification of “hot spots” for GHG emissions, and to study opportunities for precision agricultural management.

6.1. VIC-CropSyst

The effort to integrate CropSyst into a large-scale hydrologic model began with the purpose of creating a long-term Columbia River surface water supply and demand forecast encompassing municipal, agricultural, hydropower, and ecosystem concerns (Yorgey et al., 2011). The integrated model continues to undergo

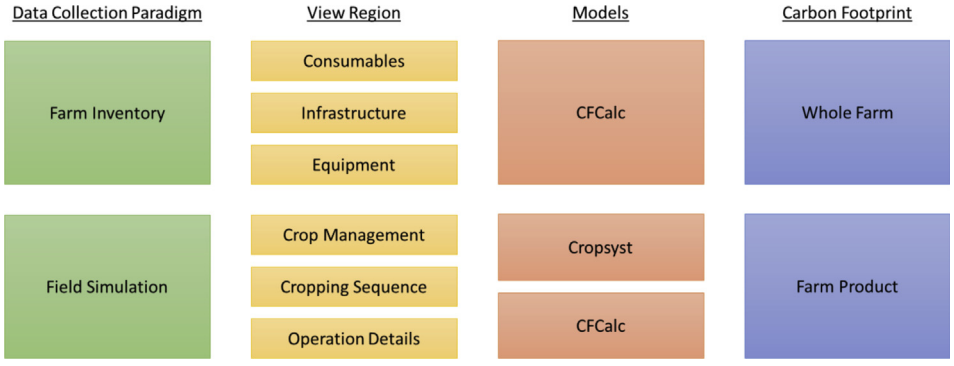


Fig. 2. A structural overview of the OFoot Tool. Regions of the website (View Region) can be grouped into two paradigms (Data Collection Paradigm) that collect data necessary to generate two levels of carbon footprints (Carbon Footprint) using subprograms (Models).

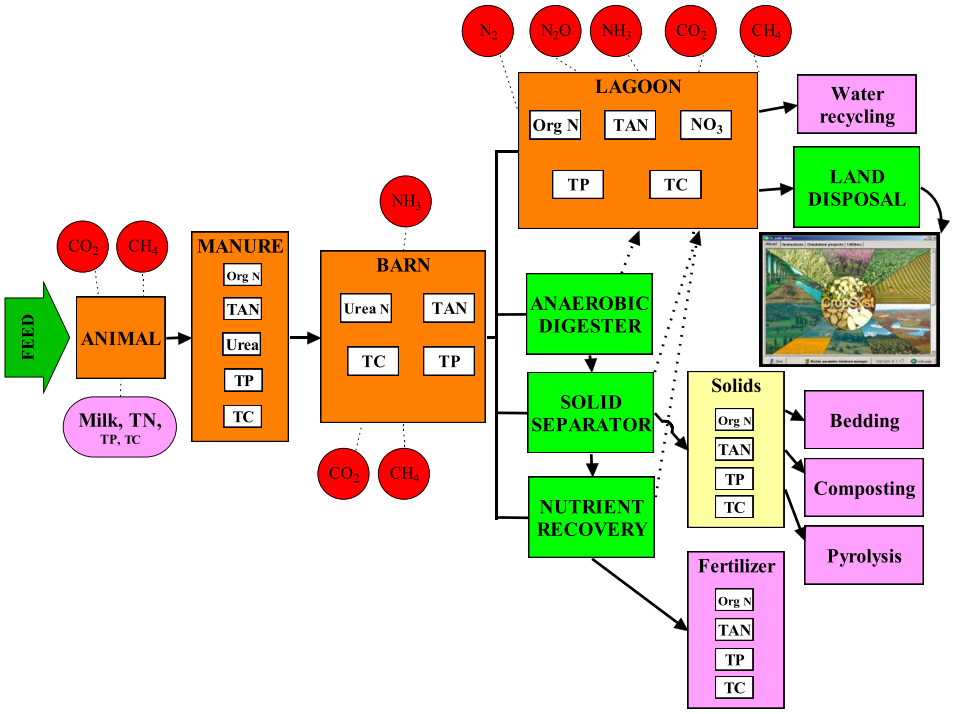


Fig. 3. Components of the integrated model for the simulation of animal nutrient management and resulting emissions and carbon footprint.

development as a central component of other large research initiatives focused on the U.S. Pacific Northwest (PNW) region. One such project involves development of the Biosphere-Relevant Earth System Model (BioEarth), a regional-scale Earth system modeling project that integrates land surfaces models with those for atmospheric, aquatic, and economic processes (Adam et al., 2014).

The Variable Infiltration Capacity (VIC; Liang et al., 1994) model is a spatially-distributed, physically-based regional-scale model which solves the water and energy budgets at every time step (from

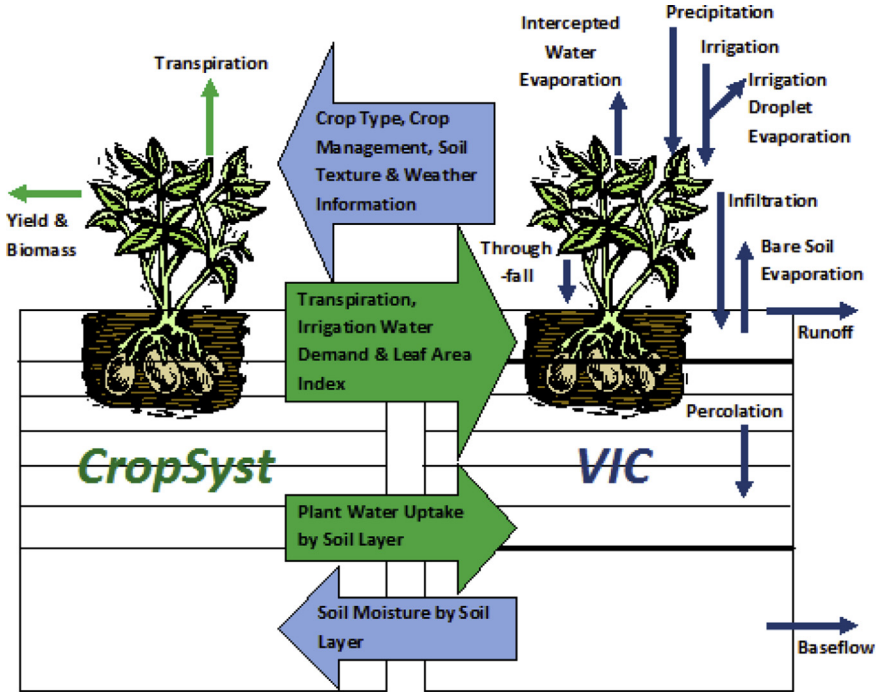


Fig. 4. VIC-CropSyst integration details. CropSyst is implemented in VIC as a dynamic vegetation and biogeochemistry sub-module, providing to VIC plant water uptake for transpiration by soil layer, water demand for irrigation, and LAI. CropSyst also simulates crop yield and biomass and carbon and nitrogen fluxes (not shown). Note: this schematic was created to emphasize the hydrologic processes that are impacted by crop growth and irrigation, and does not include a variety of other key processes, such as the snowpack, simulated by VIC.

1 to 24 h) and for every grid cell. It was developed for large-scale applications (1/16th - 2°), in which sub-grid variability in land cover and topography is based on statistical relationships. VIC accounts for key moisture and energy fluxes between the land surface and the atmosphere and includes algorithms for shallow subsurface (frozen and unfrozen) moisture, snow, lake, and wetland dynamics (Cherkauer and Lettenmaier, 1999; Andreadis et al., 2009; Bowling and Lettenmaier, 2010). VIC has been applied at multiple scales including global (e.g., Nijssen et al., 2001a,b,c; Barnett et al., 2005; Adam et al., 2009), continental U.S. (e.g., Maurer et al., 2002; Livneh et al., 2013), and over the PNW region (e.g., Hamlet and Lettenmaier, 1999; Elsner et al., 2010).

The fully-integrated VIC-CropSyst model allows simulation of the tightly-coupled dynamics between hydrologic and biogeochemical processes and producer decisions over both irrigated and dryland agricultural regions. Fig. 4 shows the integration of models, where VIC simulates the hydrologic processes with the exception of transpiration, which is provided by CropSyst. In this implementation, VIC utilizes a large number of finely-spaced soil water layers to capture dynamics in the root zone. VIC is modified to simulate crop-specific potential evapotranspiration and to represent irrigation technology-specific evaporation of irrigation water droplets in addition to bare soil and canopy-interception evaporation losses. This allows an improved representation of the amount of irrigation water that is provided to the crops versus lost through evaporation, surface runoff, or deep drainage. Also simulated by VIC-CropSyst are crop yields, and carbon and nitrogen fluxes.

The VIC model was originally developed to couple to regional and global-scale atmospheric models and is therefore rich in its representation of water and energy fluxes and has the flexibility to run at sub-daily time-steps, both of which are necessary for understanding land–atmosphere interactions and for atmospheric coupling. However, like other macroscale land surface models, VIC is coarse in its representation of lateral hydrologic processes. To reduce computation time and avoid breaking model assumptions, grids are necessarily large and sub-grid heterogeneity in land cover, elevation, saturated extent, and a variety of other variables are captured implicitly in space through semi-empirical relationships. While VIC-CropSyst does not resolve individual farms (farms are represented implicitly as sub-grid classes within a grid cell; see Fig. 5), VIC-CropSyst's strength lies in producing information relevant at regional scales when there are widespread changes in irrigation, fertilization, or crop management.

VIC-CropSyst can be applied as a component of more complex integrated modeling platforms. For example, Fig. 6 demonstrates how the model is combined with water management models to estimate the impacts of water shortages in irrigated systems. A modeling platform for the Columbia River basin (CRB) using the ColSim reservoir model (Hamlet and Lettenmaier, 1999) and spatial datasets of water rights and instream flow target requirements has been developed. This platform and others developed as part of the BioEarth project (Adam et al., 2014) are examples of how VIC-CropSyst can be applied to study the nexus of issues surrounding food, water, energy, and the climate system at larger geographic scales.

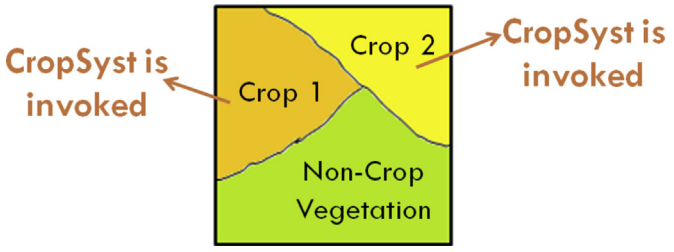


Fig. 5. An illustration of how heterogeneity in land cover is handled in VIC-CropSyst. The model is run for each of the "sub-grids" that are associated with each land cover type. These sub-grids are not explicitly located in space but are lumped together as a single unit for each grid cell. CropSyst is invoked only for the sub-grids occupied by cropland (Figure from Yorgey et al., 2011).

6.2. CropSyst Microbasin

This is a distributed hydrologic model intended for small agriculture watersheds where cropping systems management can have measurable local impacts on productivity, hydrology and nutrient cycling and flows. For example, CropSyst-Microbasin can be applied to study the impact of precision management on agriculture. The scale of analysis is over squared grid cells of a few meters in length, depending on the resolution of available digital elevation models. A prototype version of the model has been developed in MS Excel using the embedded Visual Basic for Applications programming language. The model is currently under evaluation, and a stand-alone version utilizing the distribution version of CropSyst, including parallel computing capabilities, is also under development.

The water budget in the model includes subroutines of water precipitation, irrigation and snowmelt (Brooks and Boll, 2005), water redistribution (vertical flow, overland flow and subsurface lateral flow), crop transpiration and evaporation. The model offers two options to input weather data, hourly or daily, with the latter disaggregated to hourly values during a simulation run.

Vertical infiltration can be determined using an hourly time step finite difference approach for layered soils, allowing for layers with different properties (Ross and Bristow, 1990), or using an hourly cascade method recently implemented based on concepts presented by Campbell (1985). Soil freezing and its effect on water infiltration and redistribution is also accounted for. The lateral subsurface flow follows the procedure described by Brooks et al. (2007), where lateral flow leaving a cell is calculated using Darcy's Law in short time steps (typically 180 s), with the assumption that the hydraulic gradient is equal to the land surface slope of each cell. Surface slopes also determine the flow direction into eight neighboring cells in each soil layer. The overland flow is determined using a kinematic wave approach, with outflow from one cell also divided into eight possible directions. The water depth by cell available for surface flow is determined hourly by the vertical infiltration routine.

Fig. 7 depicts the pseudo tri-dimensional water and solutes flow in CropSyst-Microbasin. CropSyst crop growth and vertical infiltration and solute transport are instantiated (i.e., a "CropSyst object" is generated) for each cell. During a simulation period, hourly runs of CropSyst and vertical flows are performed, at the end of which all state variables are "frozen" to allow for water and solute transport among cells to take place at smaller time steps, with the control returning to CropSyst objects for another hourly time step and so on until the simulation period of interest is completed.

7. Concluding remarks and future developments

The development of CropSyst during the last decade has been driven by rapidly growing application demands. This is probably similar to the experience of most cropping systems models. This article has discussed the application of CropSyst in case studies, the development and application of a grid-based framework for regional analyses, and the incorporation of CropSyst into hydrologic models, specialized decision-support tools, and multi-model

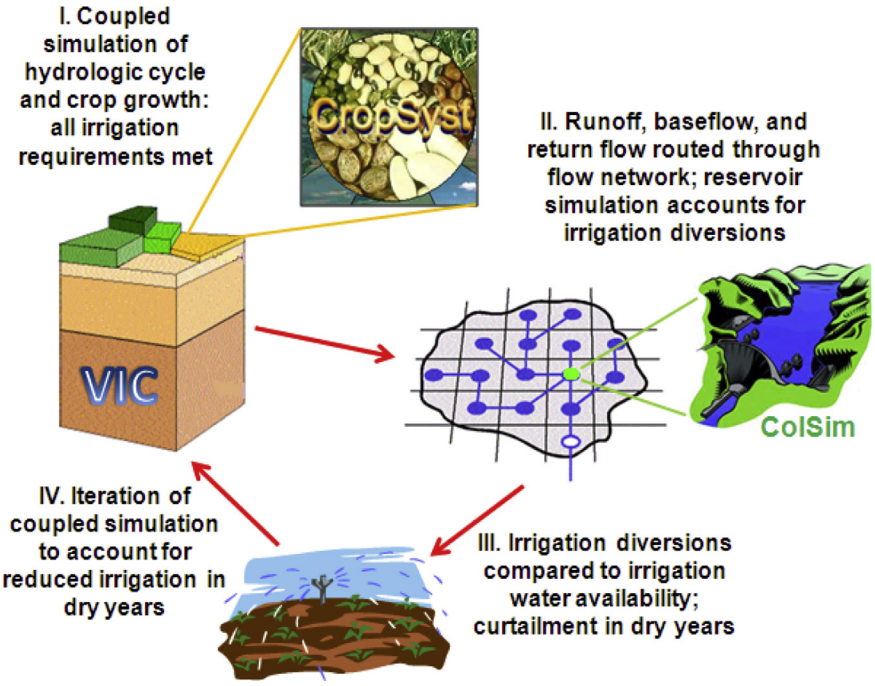
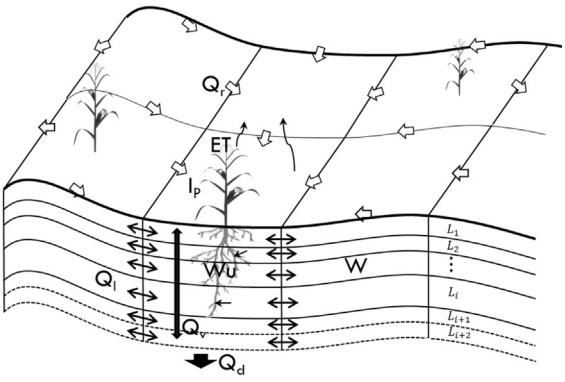


Fig. 6. VIC-CropSyst as part of an integrated modeling framework to assess the impacts of water scarcity in irrigated systems.

assessment tools. This line of work will continue and expand in the future.

Our vision for the future is to provide Internet-based open access to a library of CropSyst modules, algorithms and solutions, many of which are already available on demand.

Calls for more modularity in model development have been made to facilitate more systematic model development, documentation, maintenance, and sharing (e.g., Jones et al., 2001). Although modularity is certainly useful, an extension of this concept is the development of models as a set of standalone components that can be distributed and repackaged into different types of applications and combined with other models (e.g., hydrology, economics, bio-geochemical, and even other cropping systems models) to gain new capabilities. Indeed, there is a need to evolve from the concept of "branded models" and for the community of modelers to generate an open-source offering of components that can allow the new generation of modelers to understand previous work and reuse and modify existing solutions. An example of future directions is given by the component-oriented infrastructure provided by the European modeling platform BioMA, which enables running existing biophysical models and new modeling solutions in new modeling environments (Donatelli and Rizzoli, 2008).



Input: Precipitation (Rain and Snow) Snowmelt, Irrigation)
Flow and Storage:
ET – Evapotranspiration
I_p – Interception capacity
I – Infiltration
Wu- Water uptake
Q_r – Overland Flow
Q_v – Vertical Flow
Q_l – Lateral Flow
Q_d – Drainage
W – Water content
L – Layer of soil
Weather condition:
Daily or Hourly:
Minimum and Maximum temperature
Wind speed, Relative humidity
Solar radiation

Fig. 7. Schematic diagram of soil water flows in CropSyst-Microbasin.

Envisioned CropSyst developments will take advantage of increasing hardware, software, Internet-based and social media capabilities to create new decision-support tools using the entire model or selected modules and solutions. For example, we are currently developing a world-wide web-based scenario generation and simulation runs control for selected CropSyst applications. Also, we are adapting the crop growth routines from other cropping systems models to run within the CropSyst framework so that it will be possible to easily compare crop growth solutions under a common set of environmental conditions or run crop rotations mixing crop modules from different modeling teams.

We also envision that increased cooperation of multidisciplinary teams of scientists and a paradigm change towards more transparency, active exchange of modeling solutions, and open-source cooperative development within the global community of agricultural systems modelers will help meeting the societal demand for projections of conditions in a future world. Projects such as AgMIP (http://www.agmip.org/), MACSUR (http://www.macsur.eu/), and MODEXTREME (http://www.modextreme.org) are bringing together climate, socio-economic, agricultural and other modelers into communities with targeted plans of action. These emerging global modeling communities are providing a forum for exciting multi-disciplinary interactions and opportunities for collective model improvements as it never existed before.

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References

Abraha, M.G., Savage, M.J., 2006. Potential impacts of climate change on the grain yield of maize for the midlands of KwaZulu-Natal, South Africa. Agric. Ecosyst. Environ. 115, 150–160.

Adam, J.C., Hamlet, A.F., Lettenmaier, D.P., 2009. Implications of global climate change for snowmelt hydrology in the twenty-first century. Hydrol. Process. 23, 962–972.

Adam, J.C., Stephens, J.C., Brady, M.P., Chung, S.H., Evans, R.D., Kruger, C.E., Lamb, B.K., Liu, M.L., Stöckle, C.O., Vaughan, J.K., Rajagopalan, K., Harrison, J.A., Tague, C.L., Kalyanaramani, A., Chenj, Y., Guentherk, A., Leung, F.Y., Leung, L.R., Perleberg, A.B., Yoder, J., Allen, E., Anderson, S., Chandrasekharan, B., Malek, K., Mullis, T., Miller, C., Nergui, T., Poinsatte, J., Reyes, J., Chinnayakanahalli, K.J., Choate, M.J., Jiang, X., Nelson, R., Yoon, J.H., Yorgey, G.G., Zhu, J., Hamlet, A.F., Nijssen, B., 2014. BioEarth: a regional biosphere-relevant earth system model for understanding the consequences of agricultural and natural resource management decisions. Clim. Change (Published online: 24 April 2014).

Andreadis, K.M., Storck, P., Lettenmaier, D.P., 2009. Modeling snow accumulation and ablation processes in forested environments. Water Resour. Res. 45.

Anwar, M.R., O'Leary, G., McNeil, D., Hossain, H., Nelson, R., 2007. Climate change impact on rainfed wheat in south-eastern Australia. Field Crops Res. 104, 139–147.

Argent, R.M., 2004. An overview of model integration for environmental applications – components, frameworks and semantics. Environ. Model. Softw. 19, 219–234.

Arnold, J.G., Allen, P.M., Volk, M., Williams, J.R., Bosch, D.D., 2010. Assessment of different representations of spatial variability on SWAT model performance. Trans. ASABE 53, 1433–1443.

Asseng, S., Ewert, F., Rosenzweig, C., Jones, J.W., Hatfield, J.L., Ruane, A., Boote, K.J., Thorburn, Rötter, R.P., Cammarano, D., Brisson, N., Basso, B., Martre, P., Aggarwal, P.K., Angulo, C., Bertuzzi, P., Biernath, C., Challinor, A.J., Doltra, J., Gayler, S., Goldberg, R., Grant, R., Heng, L., Hooker, J., Hunt, L.A., Ingwersen, J., Izaurralde, R.C., Kersebaum, K.C., Müller, C., Naresh Kumar, S., Nendel, C., O'Leary, G., Olesen, J.E., Osborne, T.M., Palosuo, T., Priesack, E., Ripoche, D., Semenov, M.A., Shcherbak, I., Steduto, P., Stöckle, C., Stratonovitch, P., Streck, T., Supit, I., Tao, F., Travasso, M., Waha, K., Wallach, D., White, J.W., Williams, J.R., Wolf, J., 2013. Quantifying uncertainties in simulating wheat yields under climate change. Nat. Clim. Change 3, 827–832 (Advance Online Publication).

Barnett, T.P., Adam, J.C., Lettenmaier, D.P., 2005. Potential impacts of a warming climate on water availability in snow-dominated regions. Nature 438, 303–309.

Bechini, L., Stöckle, C.O., 2007. Integration of a cropping systems simulation model and a relational database for simple farm-scale analyses. Agron. J. 99, 1226–1237.

Bechini, L., Di Guardo, A., Botta, M., Greco, S., Maggiore, T., 2008. A simulation software for the analysis of cropping systems in livestock farms. Ital. J. Agron. 3, 213–223.

Bocchiola, D., Nana, E., Soncini, A., 2013. Impact of climate change scenarios on crop yield and water footprint of maize in the Po valley of Italy. Agric. Water Manag. 116, 50–61.

Bowling, L.C., Lettenmaier, D.P., 2010. Modeling the effects of lakes and wetlands on the water balance of Arctic environments. J. Hydrometeorol. 11, 276–295.

Brisson, N., Gary, C., Justes, E., Roche, R., Mary, B., Ripoche, D., Zimmer, D., Sierra, J., Bertuzzi, P., Burger, P., Bussiere, F., Cabidoche, Y.M., Cellier, P., Debaeke, P., Gaudillere, J.P., Henault, C., Maraux, F., Seguin, B., Sinoquet, H., 2003. An overview of the crop model STICS. Eur. J. Agron. 18, 309–332.

Brooks, E.S., Boll, J., 2005. A simple GIS-based snow accumulation and melt model. In: Proceedings of the 2005 Western Snow Conference, 11–14 April, Great Falls, MT.

Brooks, E.S., Boll, J., McDaniel, P.A., 2007. Distributed and integrated response of a geographic information system-based hydrologic model in the eastern Palouse region, Idaho. Hydrol. Process. 21, 110–122.

Campbell, G.S., 1985. Soil physics with basic: transport models for soil-plant systems. In: Developments in Soil Science, vol. 14. Elsevier, Amsterdam, 150 pp.

Cherkauer, K.A., Lettenmaier, D.P., 1999. Hydrologic effects of frozen soils in the upper Mississippi River basin. J. Geophys. Res. Atmos. 104, 19599–19610.

Confalonieri, R., Donatelli, M., Bregaglio, S., Tubiello, F.N., Fernades, E., 2012. Agro-ecological Zones Simulator (AZS): a component based, open-access, transparent platform for climate change – productivity impact assessment in Latin America. In: International Environmental Modelling and Software Society (iEMSs), 2012 International Congress on Environmental Modelling and Software, Managing Resources of a Limited Planet, Sixth Biennial Meeting, Leipzig, Germany.

Djanibekov, N., Sommer, R., Djanibekov, U., 2013. Evaluation of effects of cotton policy changes on land and water use in Uzbekistan: application of a bio-economic farm model at the level of a water users association. Agric. Syst. 118, 1–13.

Donatelli, M., Rizzoli, A.E., 2008. A design for framework-independent model components of biophysical systems. In: Sànchez-Marré, M., Béar, J., Comas, J., Rizzoli, A.E., Guariso, G. (Eds.), Proceedings of the 2008 International Congress on Environmental Modelling and Software, Barcelona, Spain, vol. 2, pp. 734–737.

Donatelli, M., Bregaglio, S., Confalonieri, R., De Mascellis, R., Acutis, M., 2014. A generic framework for evaluating hybrid models by reuse and composition – a case study on soil temperature simulation. Environ. Model. Softw. xxx, 1–9.

Elsner, M.M., Cuo, L., Voisin, N., Deems, J.S., Hamlet, A.F., Vano, J.A., Mickelson, K.E.B., Lee, S.-Y., Lettenmaier, D.P., 2010. Implications of 21st century climate change for the hydrology of Washington State. Clim. Change 102, 225–260.

Finger, R., 2012. Modeling the sensitivity of agricultural water use to price variability and climate change—an application to Swiss maize production. Agric. Water Manag. 109, 135–143.

Finger, R., Hediger, W., Schmid, S., 2011. Irrigation as adaptation strategy to climate change—a biophysical and economic appraisal for Swiss maize production. Clim. Change 105, 509–528.

Finger, R., 2013. Expanding risk consideration in integrated models - the role of downside risk aversion in irrigation decisions. Environ. Model. Softw. 43, 169–172.

Gallardo, M., Gimenez, C., Martinez-Gaitan, C., Stöckle, C., Thompson, R., Granados, M., 2012. VegSyst, a simulation model of daily crop growth, nitrogen uptake and evapotranspiration for pepper crops for use in an on-farm decision support system. Irrig. Sci. 31 (Published online February 5, 2012).

Giannakopoulos, C., Le Sager, P., Bindi, M., Moriondo, M., Kostopoulou, E., Goodess, C.M., 2009. Climatic changes and associated impacts in the Mediterranean resulting from a 2 °C global warming. Glob. Planet. Change 68, 209–224.

Hamlet, A.F., Lettenmaier, D.P., 1999. Effects of climate change on hydrology and water resources in the Columbia River Basin 1. JAWRA J. Am. Water Resour. Assoc. 35, 1597–1623.
Hartkamp, A.D., White, J.W., Hoogenboom, G., 1999. Interfacing geographic information systems with agronomic modeling: a review. Agron. J. 91, 761–772.
Jara, J., Stöckle, C.O., 1999. Simulation of corn water uptake using models with different levels of process detail. Agron. J. 91, 256–265.
Japtag, S.S., Jones, J.W., 2002. Adaptation and evaluation of the CROPGRO-soybean model to predict regional yield and production. Agric. Ecosyst. Environ. 93, 73–85.
Jones, J.W., Keating, B.A., Porter, C.H., 2001. Approaches to modular model development. Agric. Syst. 70, 412–443.
Kemanian, A.R., Stöckle, C.O., Huggins, D.R., 2005. Barley transpiration-use efficiency. Agric. For. Meteorol. 130, 1–11.
Kelly, R.A., Jakeman, A.J., Barreteau, O., Borsuk, M.E., ElSawah, S., Hamilton, S.H., Henriksen, H.J., Kuiffa, S., Maier, H.R., Rizzoli, A.E., van Delden, H., Voinov, A.A., 2013. Selecting among five common modelling approaches for integrated environmental assessment and management. Environ. Model. Softw. 47, 159–181.
Kemanian, A.R., Stöckle, C.O., Huggins, D.R., Viega, L.M., 2007. A simple method to estimate harvest index in grain crops. Field Crops Res. 103, 208–216.
Kremer, C., Stöckle, C.O., Kemanian, A.R., Howell, T., 2008. Using a canopy transpiration and photosynthesis model for the evaluation of simple models of crop productivity. In: Ahuja, L.R., Reddy, V.R., Saseendran, S.A., Yu, Q. (Eds.), Advances in Agricultural Systems Modeling 1. ASA-SSSA-CSSA, Madison, WI.
Liang, X., Lettenmaier, D.P., Wood, E.F., Burges, S.J., 1994. A simple hydrologically based model of land surface water and energy fluxes for general circulation models. J. Geophys. Res. Atmos. 99, 14415–14428.
Liu, J., 2009. A GIS-based tool for modelling large-scale crop-water relations. Environ. Model. Softw. 24, 411–422.
Livneh, B., Rosenberg, E.A., Lin, C., Nijssen, B., Mishra, V., Andreadis, K.M., Maurer, E.P., Lettenmaier, D.P., 2013. A long-term hydrologically based data set of land surface fluxes and states for the conterminous U.S.: update and extensions. J. Clim. 27, 130712132648002.
Marsal, J., Utset, A., 2008. Vineyard full-irrigation requirements under climate change scenarios for the Ebro valley, Spain. ISHS Acta Hortic. 803, 131–138.
Marsal, J., Stöckle, C., 2012. Use of CropSyst as a decision support system for scheduling regulated deficit irrigation in a pear orchard. Irrig. Sci. 30, 139–147.
Maurer, E.P., Wood, A.W., Adam, J.C., Lettenmaier, D.P., Nijssen, B., 2002. A long-term hydrologically based dataset of land surface fluxes and states for the conterminous United States*. J. Clim. 15, 3237–3251.
McCown, R.L., Hammer, G.L., Hargreaves, J.N.G., Holzworth, D.P., Freebaim, D.M., 1996. APSIM: a novel software system for model development, model testing and simulation in agricultural systems research. Agric. Syst. 50, 255–271.
Moriondo, M., Giannakopoulos, C., Bindi, M., 2011. Climate change impact assessment: the role of climate extremes in crop yield simulation. Clim. Change 104, 679–701.
Negm, L.M., Youseff, M.A., Skaggs, R.W., Chescheir, G.M., Jones, J., 2014. DRAINMOD-DSSAT model for simulating hydrology, soil carbon and nitrogen dynamics, and crop growth for drained crop land. Agric. Water Manag. 137, 30–45.
Nijssen, B., O'Donnell, G.M., Hamlet, A.F., Lettenmaier, D.P., 2001a. Hydrologic sensitivity of global rivers to climate change. Clim. Change 50, 143–175.
Nijssen, B., O'Donnell, G.M., Lettenmaier, D.P., Lohmann, D., Wood, E.F., 2001b. Predicting the discharge of global rivers. J. Clim. 14, 3307–3323.
Nijssen, B., Schnur, R., Lettenmaier, D.P., 2001c. Global retrospective estimation of soil moisture using the variable infiltration capacity land surface model, 1980–93. J. Clim. 14, 1790–1808.
Oyarzun, R.A., Stöckle, C.O., Whiting, M.D., 2007. A simple approach to modeling radiation interception by fruit-tree orchards. Agric. For. Meteorol. 142, 12–24.
Priya, S., Shibasaki, R., 2001. National spatial crop yield simulation using GIS-based crop production model. Ecol. Model. 135, 113–129.
Ross, P.J., Bristow, K.L., 1990. Simulating water movement in layered and gradational soils using the Kirchhoff transform. Soil Sci. Soc. Am. 54, 1519–1524.
Sommer, R., Glazirina, M., Yuldashev, T., Otarov, A., Ibraeva, M., Martynova, L., Bekenov, M., Kholov, B., Ibragimov, N., Kobilov, R., Karaev, S., Sultonov, M., Khasanova, F., Esanbekov, M., Mavlyanov, D., Isaev, S., Abdurahimov, S., Ikramov, R., Shezdyukova, L., de Pauw, E., 2013. Impact of climate change on wheat productivity in Central Asia. Agric. Ecosyst. Environ. 178, 78–99.
Sommer, R., Piggin, C., Haddad, A., Hajdibo, A., Hayek, P., Khalil, Y., 2012. Simulating the effects of minimum tillage and crop residue retention on water relations and yield of wheat under rainfed semiarid Mediterranean conditions. Field Crops Res. 132, 40–52.
Sommer, R., Djanibekov, N., Salaev, O., 2010. Optimization of Land and Resource Use at Farm-aggregated Level in the Aral Sea Basin of Uzbekistan with the Integrated Model FLEOM – Model Description and First Application. ZEF - Discussion Papers on Development Policy No. 139. Center for Development Research, Bonn. July 2010, 102 pp. http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1650631.
Sommer, R., Djanibekov, N., Müller, M., Salaev, O., 2011. Economic-ecological optimization model of land and resource use at farm-aggregated level. In: Martius, C., Rudenko, I., Lamers, J.P.A., Vlek, P.L.G. (Eds.), Cotton, Water, Salts and Soums - Economic and Ecological Restructuring in Khorezm, Uzbekistan. Springer New York, LLC, pp. 267–283, 419 pp.
Stöckle, C.O., Debaeke, P., 1997. Modelling crop nitrogen requirements: a critical analysis. Eur. J. Agron. 7, 161–169.
Stöckle, C.O., Donatelli, M., Nelson, R., 2003. CropSyst, a cropping systems simulation model. Eur. J. Agron. 18, 289–307.
Stöckle, C.O., Nelson, R.L., Higgins, S., Brunner, J., Grove, G., Boydston, R., Whiting, M., Kruger, C., 2010. Assessment of climate change impact on eastern Washington agriculture. Clim. Change 102, 77–102.
Stöckle, C., Higgins, S., Kemanian, A., Nelson, R., Huggins, D., Marcos, J., Collins, H., 2012. Carbon storage and nitrous oxide emissions of cropping systems in eastern Washington: a simulation study. J. Soil Water Conserv. 67 (5), 365–377.
Tingem, M., Rivington, M., 2009. Adaptation for crop agriculture to climate change in Cameroon: tuning on the heat. Mitig. Adapt. Strateg. Glob. Change 14, 153–168.
Torriani, D.S., Calanca, P., Schmid, S., Beniston, M., Fuhrer, J., 2007. Potential effects of changes in mean climate and climate variability on the yield of winter and spring crops in Switzerland. Clim. Res. 34, 59–69.
Tripathy, R., Ray, S.S., Kaur, H., Jalota, S.K., Bal, S.K., Panigrahy, S., 2011. Understanding spatial variability of cropping system response to climate change in Punjab State of India using remote sensing data and simulation model. In: International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, vol. XXXVIII-8/W20. ISPRS Bhopal 2011 Workshop, 8 November 2011, Bhopal, India.
Tubiello, F.N., Donatelli, M., Rosenzweig, C., Stöckle, C.O., 2000. Effects of climate change and elevated CO2 on cropping systems: model predictions at two Italian locations. Eur. J. Agron. 13, 179–189.
Uslar, N., 2010. Gaseous Emissions and Nutrient Fate Modeling in Dairy Operations (Ph.D. dissertation). Washington State University, Pullman, WA 99164, 111 pp.
Van Ittersum, M., Donatelli, M., 2003. Modelling cropping systems – highlights of the symposium and preface to the special issues. Eur. J. Agron. 18, 187–197.
Williams, J.R., Jones, C.A., Dyke, P.T., 1984. A modeling approach to determining the relationship between erosion and soil productivity. Trans. ASAE 27, 129–144.
Yorgey, G.G., Rajagopalan, K., Chinnayakanahalli, K., Brady, M.P., Barber, M.E., Nelson, R., Stöckle, C.S., Kruger, C.E., Dinesh, S., Malek, K., Yoder, J., Adam, J.C., 2011. Columbia River Basin Long-term Water Supply and Demand Forecast. Washington State Legislative Report. http://www.ecy.wa.gov/biblio/1112011.html.
Zaher, U., Stöckle, C.O., Painter, K., Higgins, S., 2013. Life cycle assessment of the potential carbon credit from no- and reduced-tillage winter wheat-based cropping systems in eastern Washington State. Agric. Syst. 122, 73–78 (on line).